

More on determinants

Theorem 4. Let A be an $n \times n$ matrix

- (a) If matrix B is obtained from A by multiplying a row (or column) of A by a constant λ , $\det(B) = \lambda \det(A)$. We then have $\det(\lambda A) = \lambda^n \det(A)$.
- (b) If matrix B is obtained from A by adding a multiple of one row of A to a different row of A , then $\det(B) = \det(A)$.

Proof. (a) Let B be the matrix obtained by multiplying the i -th row of A by λ , then B has form

$$B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{i1} & \lambda a_{i2} & \cdots & \lambda a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

expanding along the i -th row we have $\det(B) = \sum_{j=1}^n (\lambda a_{ij}) C_{ij} = \lambda \sum_{j=1}^n a_{ij} C_{ij} = \lambda \det(A)$. $\det(\lambda A) =$

$\lambda^n \det(A)$ follows from this (make sure you know why as discussed in class).

(b) Let B be a matrix obtained by adding λ multiple of j -th row to the i -th row, then B has form

$$B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} + \lambda a_{j1} & a_{i2} + \lambda a_{j2} & \cdots & a_{in} + \lambda a_{jn} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

expanding $\det(B)$ along the i -th row we get $\det(B) = \sum_{k=1}^n (a_{ik} + \lambda a_{jk}) C_{ik} = \sum_{k=1}^n a_{ik} C_{ik} + \lambda \sum_{k=1}^n a_{jk} C_{ik} = \det(A) + \lambda \det(C)$ where C is the matrix obtained from A by replacing row i by row j , since C has two identical rows, then $\det(C) = 0$ and so $\det(B) = \det(A)$ $\square$

Examples (1) Let A be a 4×4 matrix such that $\det(A) = \pi$, then $\det(10A^T) = 10^4 \det(A^T) = 10^4 \det(A) = 10^4 \pi$

(2) If we want to calculate determinant of matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 1 \\ 2 & 0 & 1 \end{pmatrix}$, we could proceed by reducing to a simpler matrix with same determinant and expand by the first column

$$\begin{aligned} \det(A) &= \begin{vmatrix} 1 & 2 & 3 \\ 1 & 4 & 1 \\ 2 & 0 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 2 & 3 \\ 0 & 2 & -2 \\ 2 & 0 & 1 \end{vmatrix} \quad (R_2 \rightarrow R_2 - R_1) \\ &= \begin{vmatrix} 1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & -4 & -5 \end{vmatrix} \quad (R_3 \rightarrow R_3 - 2R_1) \\ &= 1 \begin{vmatrix} 2 & -2 \\ -4 & -5 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 \\ -4 & -5 \end{vmatrix} + 0 \begin{vmatrix} 2 & 3 \\ 2 & -2 \end{vmatrix} \\ &= -18 \end{aligned}$$

As remarked in class, be careful with your row operations in the second example as it might change your determinant.

Theorem 5. Let E be an $n \times n$ elementary matrix, then

- (a) If E is associated with adding a multiple of one row to a different row, then $\det(E) = 1$.
- (b) If E is associated with interchanging two distinct rows, then $\det(E) = -1$.
- (c) If E is associated with multiplying the i -th row by a non-zero scalar λ , then $\det(A) = \lambda$.
- (d) If A is any $n \times n$ matrix, then $\det(EA) = \det(E)\det(A)$.

Proof. (a) E will either be lower or upper triangular with 1's on the diagonal, and so $\det(E) = 1$ (or use above results $\det(E) = \det(I_n) = 1$).

(b) E is obtained by interchanging two different rows of the identity matrix and so $\det(E) = -\det(I) = -1$.

(c) E is diagonal matrix with $\det(E) = \lambda$ (or use above results $\det(E) = \lambda \det(I_n) = \lambda$).

(d) Check for each case as discussed in class. □

Theorem 6. Let A be an $n \times n$ matrix, then A is invertible if and only if $\det(A) \neq 0$.

Proof. Suppose A is invertible, then there exists elementary matrices $E_1, E_2, \dots, E_k$ such that $A = E_k \cdots E_2 E_1$, note that $\det(A) = \det(E_k) \cdots \det(E_2) \det(E_1)$, and since we have seen that determinants of elementary matrices are non-zero, we have that $\det(A) \neq 0$, conversely, suppose $\det(A) \neq 0$, we know that we can always row reduce an $n \times n$ matrix A to a matrix U in row echelon form, we thus have elementary matrices $E_1, E_2, \dots, E_k$ such that $A = E_k \cdots E_2 E_1 U$ and so $U = E_1^{-1} E_2^{-1} \cdots E_k^{-1} A$, therefore $\det(U) = \det(E_1^{-1}) \det(E_2^{-1}) \cdots \det(E_k^{-1}) \det(A)$, since $\det(A) \neq 0$, we must also have that $\det(U) \neq 0$, matrix U is in row echelon form so it must be of form

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ 0 & 0 & \cdots & u_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}$$

U is upper triangular and so $\det(U) = u_{11}u_{22}\cdots u_{nn}$, then from $\det(U) \neq 0$ we must have that for each $1 \leq i \leq n$, $u_{ii} \neq 0$ and therefore U is row equivalent to the identity matrix I_n , from which we have that A is also row equivalent to I_n and thus invertible. $\square$

Eigenvalues and eigenvectors

Definition 26. Let A be an $n \times n$ matrix. A scalar λ is called an eigenvalue of matrix A if there is a vector $\mathbf{x} \neq \mathbf{0}$ such that $A\mathbf{x} = \lambda\mathbf{x}$, this non-zero vector $\mathbf{x}$ is called an eigenvector corresponding to the eigenvalue λ .

Determinants provide a useful tool to find eigenvalues of a matrix.
Let us continue the story in the next lecture...